

Explainable AutoML-driven surface water quality classification with key indicators identification

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ABSTRACT

Surface water quality monitoring and classification face dual challenges: prohibitive costs of comprehensive indicators monitoring and the labor-intensive process of traditional machine learning (ML) implementation, which jointly constrain the development of efficient water quality assessment systems. This study presents an automated and interpretable framework: Auto-sklearn automates the machine learning workflow to address model selection complexity, while SHAP analysis quantifies feature contributions to reduce water quality indicator requirements. Using monitoring data from major river basins of China, we evaluated Auto-sklearn against conventional ML algorithms and modern ensemble methods. Auto-sklearn achieved the highest mean performance (Weighted F1: 0.9633 ± 0.0027) via an optimized ensemble. Crucially, this high performance proved to be robust and generalizable, as the framework maintained its superiority across rigorous temporal and spatial validation schemes. Through Kernel SHAP analysis, we identified COD_{Mn} , TP, and DO as the key indicators. A 3-indicator Auto-sklearn model achieved a high Weighted F1 (0.9205 ± 0.0097) and outperformed the rule-based baseline, with higher F1-scores across all classes and an 18.9 % relative improvement for Class WV water. This suggests that monitoring costs can be substantially reduced while simultaneously improving assessment reliability. Further regional analysis revealed spatial variations, with most basins aligning with the nationwide set, but the Song-Liao and Yellow River Basins also requiring $\text{NH}_3\text{-N}$ monitoring. This framework provides a practical, cost-effective solution for water environment management, especially in developing regions.

1. Introduction

Water quality monitoring and early warning systems play a crucial role in environmental protection and public health, particularly in developing countries facing rapid industrialization and urbanization (Michalak, 2016; Greenhill et al., 2024; Qian et al., 2024). While the United Nations has emphasized water quality management in its Sustainable Development Goals (SDGs 6 and 14), implementing effective monitoring systems remains challenging due to technical and resource constraints (Herrera, 2019; Byrne, 2024). Surface water quality classification based on monitoring indicators is fundamental to sustainable water environment management, yet achieving both cost-effectiveness

and reliability in such classification systems remains a significant challenge in developing regions (Baccour et al., 2024).

Automatic water quality monitoring stations started to be deployed in major river basins in China since 1999, implementing classification based on the single-factor evaluation method (Table S1). This evaluation method requires monitoring of up to 24 water quality indicators. The list includes not only conventional parameters but also numerous indicators such as heavy metals (e.g., mercury, lead) and specific organic pollutants (e.g., volatile phenols), which are notoriously difficult to monitor continuously in the field. This comprehensiveness poses significant practical challenges. Specifically, sensors for pollutants such as heavy metals and organics are expensive and less robust, leading to high

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operational costs and frequent data interruptions (Durgun, 2024). As a direct consequence of these operational and financial constraints, the national online monitoring network of China pragmatically focuses on a core subset of more feasible indicators rather than the full 24 indicators. This reality highlights the urgent need to identify the most critical indicators even within this reduced set to further enhance monitoring efficiency and cost-effectiveness (Othman et al., 2020; Wu et al., 2023).

Traditional statistical approaches, such as principal component analysis and Pearson correlation analysis, have been widely used for water quality indicator selection (Tripathi and Singal, 2019; Uddin et al., 2021; Asadollah et al., 2021). With the advancement of artificial intelligence, machine learning (ML) methods have demonstrated superior capability in handling complex nonlinear relationships in water quality prediction and key indicator selection (Zhu et al., 2022; Ding et al., 2023; Zhang et al., 2024). However, developing an optimal and robust predictive system using ML presents significant challenges. The process involves a complex, multi-stage workflow, including systematically selecting the most suitable algorithm, performing robust feature engineering, and conducting rigorous, computationally intensive hyperparameter optimization (HPO). While a single model might work efficiently with default parameters, the HPO process, which systematically evaluates hundreds or thousands of configurations to maximize predictive accuracy, is often computationally prohibitive, demanding substantial resources and specialized expertise (Peng et al., 2022; Xia and Zeng, 2022; Li et al., 2024). To address these specific workflow challenges, automated machine learning (AutoML) has emerged as a powerful paradigm. It is designed to automate this entire end-to-end pipeline, from data preprocessing and feature selection to model choice and HPO, thereby greatly reducing the demand for domain-specific expertise and computational oversight (Zöller and Huber, 2021). Several AutoML methods have been developed, such as TPOT, H2O, and Auto-sklearn (Feurer et al., 2015; Olson et al., 2016; LeDell and Poirier, 2020). Among them, Auto-sklearn has demonstrated superior performance in multiple competitions through its integration of Bayesian optimization, meta-learning, and ensemble construction. Even in specially designed “Beat Auto-sklearn” challenges, manual parameter tuning by experienced data scientists could not surpass its automated optimization performance (Guyon et al., 2019).

However, the automated model selection and ensemble processes of AutoML intensify the “black box” nature of models, challenging the understanding and interpretation of model decision processes (Barredo Arrieta et al., 2020). This is particularly crucial in water quality management, where models must maintain both prediction accuracy and transparency in decision-making. The Kernel SHAP (SHapley Additive exPlanations) method offers a promising solution by combining LIME (Local Interpretable Model-agnostic Explanations) with Shapley values to construct a unified interpretation framework. Through quantitative feature importance attribution derived from cooperative game theory, SHAP effectively measures the contribution of features to model predictions (Lundberg and Lee, 2017a; Russo et al., 2023). While SHAP values show great potential, their application in water quality indicator screening remains unexplored, especially as an alternative to computationally prohibitive “best subset” selection methods. These brute-force approaches test every possible feature combination to guarantee finding the optimal set, but their computational cost grows exponentially, making them impractical for larger feature sets (Chen et al., 2020).

Herein, this study develops an automated and interpretable framework for water quality classification using monitoring data from the China National Environmental Monitoring Center (CNEMC). The framework classifies surface water quality into a six-level system (Class I-V and worse than V) based on the nine core indicators reliably monitored by the national online monitoring network of China. The specific objectives are to: (i) evaluate the performance of Auto-sklearn against conventional ML algorithms and modern ensemble methods through comprehensive metrics (Precision, Recall, Weighted F1, Macro F1,

confusion matrix, and AUC-ROC curves); (ii) interpret the Auto-sklearn ensemble model using Kernel SHAP to quantify indicator contributions to model predictions; and (iii) identify key water quality indicators for optimizing monitoring schemes. Through this integrated approach, we aim to establish a cost-effective framework that can enhance water quality management and decision-making processes while simultaneously improving assessment reliability.

2. Materials and methods

2.1. Study area and water quality dataset

Datasets of surface water quality were collected from the CNEMC, covering nine major river basins, including Continental, Hai, Huai, Pearl, Song-Liao, Southeast, Southwest, Yangtze, and Yellow rivers. The data were collected from approximately 1900 monitoring sites from January 2021 to June 2024, with a time step of 4 h (Fig. 1a). The national online monitoring system of CNEMC focuses on a core set of 11 feasible indicators for real-time tracking. However, after assessing data quality, two of these indicators [chlorophyll-a (Chl-a, mg/L) and algal density (cells/L)] were excluded from our study due to severe data missingness (>70 %, Table S3). The final analysis was therefore conducted on the remaining nine robustly-monitored indicators: water temperature (WT, °C), pH, dissolved oxygen (DO, mg/L), electrical conductivity (EC, $\mu\text{S}/\text{cm}$), chemical oxygen demand (COD_{Mn} , mg/L), ammonia nitrogen ($\text{NH}_3\text{-N}$, mg/L), total phosphorus (TP, mg/L), total nitrogen (TN, mg/L), and turbidity (NTU).

According to the Environmental Quality Standards for Surface Water (MEEPRC, 2002), water quality was classified into six classes from best to worst (I–V). This classification serves as the target variable for our supervised learning models. The classification for each data record was pre-determined by the CNEMC based on the “single-factor evaluation method”, where the final class is assigned based on the poorest-performing individual indicator relative to the standard limits as shown in Table S1. Our objective is therefore to predict this official classification using the nine selected water quality indicators as input features.

To ensure computational feasibility for the extensive automated model search and subsequent SHAP analysis, a subset of 50,000 samples was extracted from the cleaned dataset ($n = 10,159,876$) using stratified random sampling. This sampling approach was chosen to prioritize model stability, as our analysis showed that training on the full dataset produced unstable, high-variance results and required computationally prohibitive runtimes (Fig. S1). To rigorously evaluate the stability and robustness of our findings against sampling variability, this entire sampling process was repeated five times using different random seeds (42, 123, 777, 1024, and 2048).

This sampling methodology not only preserved the proportional distribution of the water quality classes (Fig. 1b) but also maintained a broadly consistent distribution across the major river basins. As detailed in Table S2, the proportional representation of each basin in the sampled subsets shows strong alignment with the full dataset, exhibiting only minor deviations, thereby confirming that the sampling process did not introduce significant regional bias. For the regional analysis, the same stratified sampling approach was applied to each major basin separately. Spearman correlation analysis was performed on the stratified dataset, and indicated no significant relationships among the selected water quality indicators (Fig. 1c), thus eliminating the need for removal of multicollinearity features.

2.2. Model development and validation strategies

For model inputs, all nine water quality indicators (WT, pH, DO, EC, COD_{Mn} , $\text{NH}_3\text{-N}$, TP, TN, and NTU) were used as input features, with water quality classes (I–V) as the output target. To rigorously evaluate model performance and stability, three distinct validation schemes were

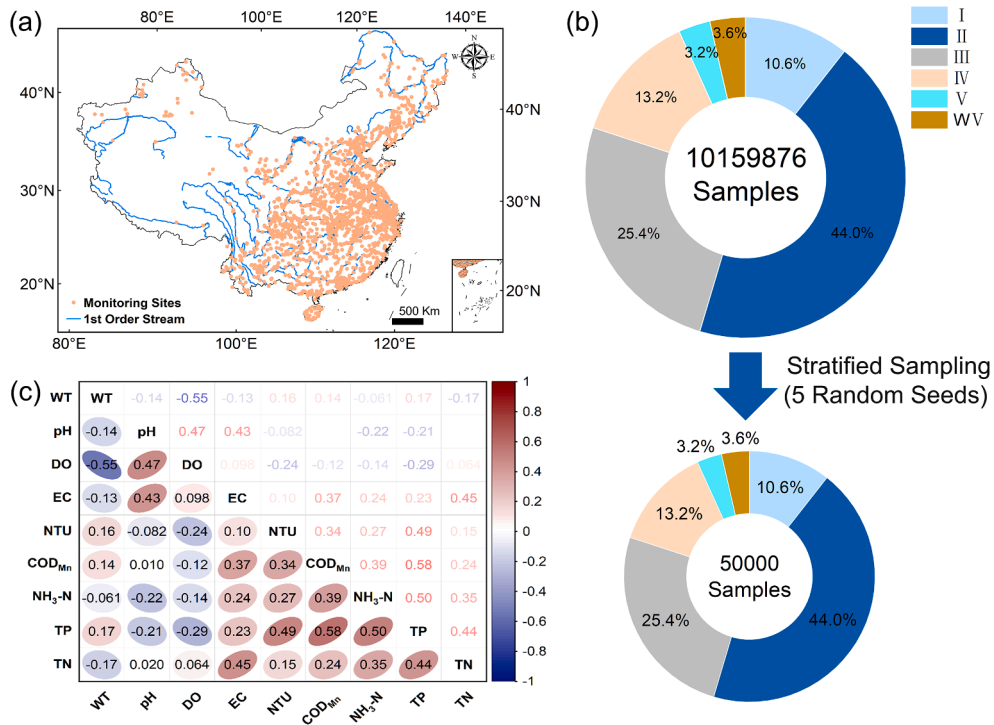


Fig. 1. Study area and dataset: (a) distribution of water quality monitoring sites in the study area; (b) distribution of water quality classes before and after stratified sampling, a process repeated five times with unique random seeds; (c) Spearman correlation heatmap of water quality indicators.

employed.

- i. Stratified random split: This was the baseline validation method. For each of the five runs, the 50,000-sample subset was randomly

partitioned into a training set (80 %, $n = 40,000$) and a test set (20 %, $n = 10,000$) while maintaining class proportions.

- ii. Temporal hold-out validation: To assess the models' ability to perform on future, unseen data, the entire clean dataset ($n = 10,159,876$) was divided based on a timestamp of "2024-01-01

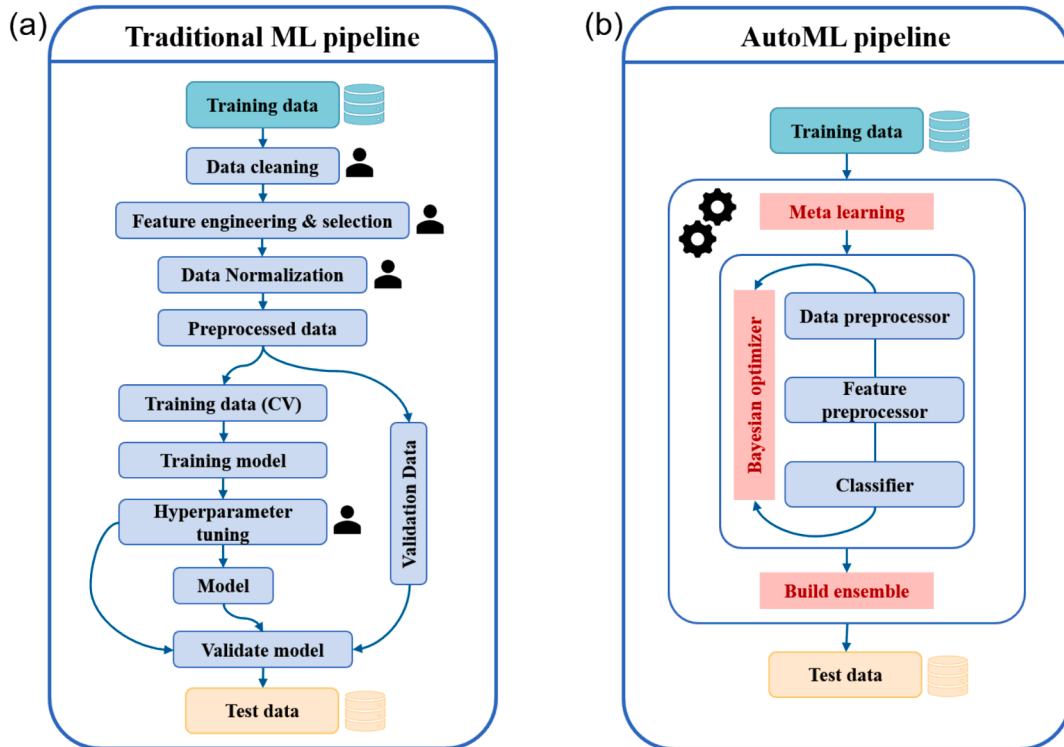


Fig. 2. Comparison of ML pipelines: (a) traditional ML pipeline requiring manual intervention at multiple stages; (b) AutoML pipeline utilizing Auto-sklearn framework, which automates the model development process through *meta*-learning, Bayesian optimization, and ensemble construction.

00:00:00". The training set ($n = 40,000$) was then sampled exclusively from data recorded before this timestamp, while the test set ($n = 10,000$) was sampled from data recorded after this point. This process was also repeated five times using different random seeds.

- iii. Spatial hold-out validation: Finally, to evaluate geographic generalizability, a leave-one-basin-out validation was conducted. This process consisted of nine iterations where each major river basin was sequentially designated as the hold-out basin. For each iteration, a training set ($n = 40,000$) was sampled from the remaining eight basins, and a test set ($n = 10,000$) was sampled from the single hold-out basin.

Consequently, all quantitative performance metrics reported in this study are presented as the mean $\pm$ Standard Deviation (SD) across the respective runs or hold-out basins to ensure the robustness of our conclusions.

2.3. Traditional and automated machine learning

Traditional ML pipelines require substantial human intervention across multiple stages, including data preprocessing, feature engineering, model selection, and hyperparameter tuning (Fig. 2a). For traditional ML models, each input feature was standardized using Z-score normalization ($\mu = 0$, $\sigma = 1$) to ensure all variables had comparable scales, as illustrated in Fig. S2. And Gaussian process-based Bayesian optimization was employed for hyperparameter tuning in traditional ML approaches (Snoek et al., 2012).

In contrast, for AutoML implementation, raw data was used without any preprocessing, allowing the algorithm to automatically determine the optimal preprocessing methods (Fig. 2b). The Auto-sklearn (version 0.15.0) framework combines Bayesian optimization, meta-learning, and ensemble construction for AutoML pipeline selection (Feurer et al., 2015). This framework incorporates comprehensive ML tools, including classification algorithms, feature preprocessing methods, and data preprocessing techniques. No manual adjustments were made to the model selection process, with Auto-sklearn automatically defining the search space for hyperparameter optimization. For the accurate assessment of model performance, due to the highly imbalanced distribution of water quality classes in the dataset, both traditional ML and AutoML utilized 5-fold stratified cross-validation during model training (Fig. S3).

For comprehensive evaluation, Auto-sklearn was compared with both conventional ML models implemented through scikit-learn [Logistic Regression (LR), k-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF)] and state-of-the-art ensemble learning models [CatBoost (CatB), LightGBM (LGBM), XGBoost (XGB)]. All implementations were conducted using Python 3.8.13. Detailed information about traditional ML methods is provided in the Supplementary Materials (Text S1).

2.4. Model evaluation metrics

The predictive performance of the models was assessed through a comprehensive suite of multi-dimensional evaluation metrics. These metrics, including their mathematical formulas and detailed statistical interpretations, are summarized in Table S4. For this multi-class classification task, we employed key metrics such as Precision, Recall, and the Weighted F1, which effectively accounts for class imbalance. Additionally, confusion matrices were used for a visual breakdown of classification performance across all classes, while Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC) values were used to evaluate the overall discriminative ability of the models.

2.5. Explanation of Auto-sklearn

To interpret the ensemble model generated by Auto-sklearn, we employed the SHapley Additive exPlanations (SHAP) method, which calculates feature importance through game theory principles (Kuhn, 2020). In ML context, Shapley values measure the individual impact of each feature in the model, collectively contributing to the prediction. Various versions of SHAP have been developed by Lundberg and Lee, including DeepSHAP, Kernel SHAP, LinearSHAP, and TreeSHAP, each designed to approximate SHAP values for different types of ML models. A critical prerequisite for a meaningful SHAP analysis is the high predictive accuracy of the model being explained. The SHAP interpretation was therefore performed only after the high predictive performance of the Auto-sklearn ensemble had been established.

Auto-sklearn can produce heterogeneous ensembles of different algorithm types. Therefore, the model-agnostic Kernel SHAP was selected for interpretation, as it is well-suited for such ensembles and requires no model-specific adjustments. It approximates SHAP values through a weighted local regression that bridges Model-agnostic Explanations (LIME) methods with Shapley values from cooperative game theory (Lundberg and Lee, 2017b). The SHAP value ϕ_i for feature i is calculated as follows:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \quad (1)$$

where $f_x(S)$ is defined as:

$$f_x(S) = E[f(x)|x_s] \quad (2)$$

where S represents the subset of features excluding feature i , F is the set of all features, M is the total number of features, and $f_x(S)$ represents the expected model prediction when only features in subset S are known. Kernel SHAP approximates these values by sampling feature coalitions and computing model predictions, making it particularly suitable for ensemble models where direct interpretation of model internals is challenging (Moradi and Samwald, 2021).

3. Results

3.1. Runtime optimization and ensemble configuration of Auto-sklearn for water quality classification

To optimize performance of Auto-sklearn for water quality classification, the impact of different training durations ranging from 0.5 to 12 h was investigated while maintaining a constant data partitioning ratio of 0.8 (training dataset = 40,000, testing dataset = 10,000). Fig. 3 presents the trend of Weighted F1 and relative improvements across different runtime periods. The Weighted F1 was chosen as the primary evaluation metric because it provides a balanced assessment of model performance by combining both Precision and Recall, being particularly suitable for evaluating the overall effectiveness of the model (detailed results for Precision and Recall are provided in Table S5). The Weighted F1 exhibited a substantial and statistically significant improvement during the initial 4 h of training, increasing from 0.9537 ± 0.0002 (0.5 h) to 0.9627 ± 0.0002 (4 h). A one-way ANOVA followed by Tukey's post-hoc test confirmed that the performance at 0.5 h, 1 h, 2 h, and 4 h were all significantly different from each other ($p < 0.05$). The high baseline performance reflects the strong, inherent relationship between the input indicators and the target classes, which are determined by these same indicators according to the national standard (Section 2.1).

With the increase in training time, a trend of diminishing returns was observed. Crucially, while the Weighted F1 increased to 0.9631 ± 0.0002 after 8 h, this improvement was not statistically significant compared to the 4-hour result ($p > 0.05$). The mean Weighted F1 increase dropped from 0.0040 (1–2 h) to 0.0029 (2–4 h), and further

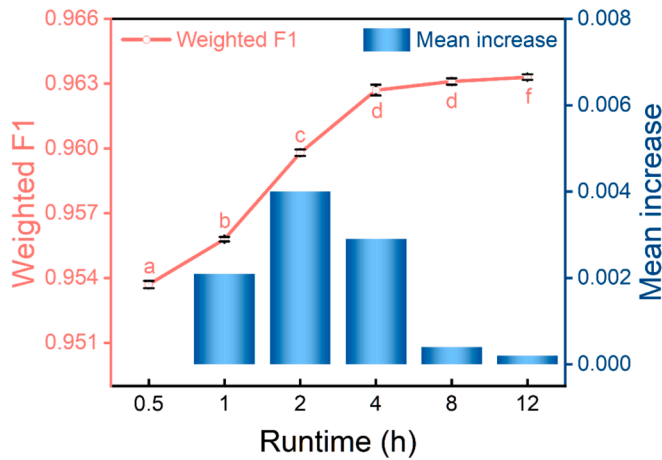


Fig. 3. Performance improvement of Auto-sklearn in water quality classification with different training runtimes. The red line shows the Weighted F1, and the blue bars indicate the mean increase compared to the previous time point. Different letters denote statistically significant differences among the runtimes (Tukey's HSD, $p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

decreased to just 0.0004 for the 4–8 h interval. Although a further significant improvement was noted at 12 h, the marginal gain was minimal relative to the extensive additional computation time. Therefore, based on these results, the 4-hour training period was selected as the optimal parameter for subsequent studies, as it strikes an effective balance between model performance and computational cost.

Through automated model selection and optimization, Auto-sklearn generated a final ensemble model consisting of 7 ML pipelines. The detailed configurations of this ensemble are now provided in the [Supplementary Materials \(Table S6\)](#). In summary, the automated process consistently selected RF as the optimal classifier across all pipelines, applying various data preprocessing and distribution-based balancing strategies to optimize performance. This strong preference for RF classifiers suggests that tree-based ensemble methods with moderate complexity are particularly effective for this water quality classification task.

3.2. Water quality classification performance and generalization

3.2.1. Comparative performance of models

The performance of Auto-sklearn was comprehensively evaluated against seven machine learning models. Based on five independent runs using a stratified random split, a Tukey's HSD statistical test ($p < 0.05$) grouped the models into distinct performance tiers. As shown in [Table 1](#), Auto-sklearn, RF, and CatB constituted the top tier, with no statistically significant performance difference among them. Within this top group, however, Auto-sklearn consistently achieved the highest mean scores

Table 1

Benchmarking results for water quality classification models, presented as mean $\pm$ SD.

	Precision	Recall	Weighted F1
Auto-sklearn	0.9636 $\pm$ 0.0027 ^a	0.9635 $\pm$ 0.0027 ^a	0.9633 $\pm$ 0.0027 ^a
RF	0.9612 $\pm$ 0.0028 ^a	0.9610 $\pm$ 0.0029 ^a	0.9607 $\pm$ 0.0029 ^a
CatB	0.9597 $\pm$ 0.0027 ^{ab}	0.9599 $\pm$ 0.0026 ^a	0.9596 $\pm$ 0.0027 ^a
DT	0.9571 $\pm$ 0.0024 ^{bc}	0.9570 $\pm$ 0.0024 ^b	0.9566 $\pm$ 0.0024 ^b
LGBM	0.9560 $\pm$ 0.0034 ^c	0.9564 $\pm$ 0.0034 ^b	0.9560 $\pm$ 0.0035 ^b
XGB	0.9546 $\pm$ 0.0030 ^c	0.9549 $\pm$ 0.0031 ^b	0.9545 $\pm$ 0.0030 ^b
KNN	0.8509 $\pm$ 0.0023 ^d	0.8523 $\pm$ 0.0023 ^c	0.8492 $\pm$ 0.0023 ^c
LR	0.6503 $\pm$ 0.0025 ^e	0.6787 $\pm$ 0.0023 ^d	0.6457 $\pm$ 0.0026 ^d

Note: For each column, different letters in superscript indicate a statistically significant difference between models (Tukey's HSD, $p < 0.05$).

across all evaluation metrics, with a precision of 0.9636 ± 0.0027 , recall of 0.9635 ± 0.0027 , and a weighted F1-score of 0.9633 ± 0.0027 . A detailed, per-class analysis presented in [Table S7](#) further underscores this leading performance, showing that Auto-sklearn also attained the highest Macro F1-score. The remaining models were grouped into subsequent performance tiers, with KNN and LR performing significantly poorer than all other methods.

The classification performance is further detailed in the confusion matrices ([Fig. 4](#)). Among the evaluated models, Auto-sklearn ([Fig. 4a](#)) emerged as the top-performing solution, demonstrating exceptional and well-balanced accuracy across all water quality classes. It achieved near-perfect scores for Class I (99.1 %) and Class II (98.4 %) and maintained high accuracy on the ambiguous intermediate classes: 96.1 % for Class III, 91.0 % for Class IV, and 84.4 % for Class V. This capability starkly contrasts with traditional models like LR ([Fig. 4d](#)), whose accuracy collapsed to just 53.3 % for Class III, 30.8 % for Class IV, and even 0 % for Class V, highlighting the severe limitations of linear models. The confusion matrices for other models are presented in [Fig. S4](#).

The Receiver Operating Characteristic (ROC) curves and corresponding Area Under the Curve (AUC) values in [Fig. 5](#) further validate the models' discriminative ability. Auto-sklearn ([Fig. 5a](#)) achieved the highest macro-average AUC (MA-AUC) of 0.9963, followed closely by other top-tier models like RF (MA-AUC = 0.9961) and CatB (MA-AUC = 0.9960). In contrast, LR ([Fig. 5d](#)) was substantially weaker with a MA-AUC of 0.9027, showing particular difficulty with Class III (AUC = 0.8257) and Class IV (AUC = 0.8762). Additional ROC analyses for the remaining models are available in [Fig. S5](#).

This comprehensive comparison underscores the ability of the Auto-sklearn framework to automatically construct a model that achieves superior water quality classification performance.

3.2.2. Generalization assessment with temporal and spatial validation

To more rigorously assess model robustness and address potential data leakage from simple random splits, two additional validation schemes were implemented: a temporal hold-out (training on data before Jan 1, 2024, and testing on data after) and a spatial hold-out (leave-one-basin-out cross-validation). The results are summarized in [Table 2](#).

The temporal validation results demonstrate the model's robust classification performance, which remains stable across different time periods. The performance of models remained remarkably stable, with no statistically significant drop in the Weighted F1-score when predicting on future, unseen data. This indicates that the models effectively learned underlying patterns that are consistent over time.

The spatial validation provided critical insights into the geographic generalizability of models. While the average performance across the basin-level cross-validation remained high, the standard deviation increased significantly—for Auto-sklearn, from 0.0027 in the random split to 0.0154. This high variance is strong quantitative evidence of underlying regional heterogeneity, indicating that the model's performance, while strong overall, varies across basins with distinct characteristics ([Table S8](#)). Crucially, despite this performance variability, the relative ranking of the models remained consistent. This stability in relative performance confirms that the Auto-sklearn framework is a robust and reliable choice for nationwide water quality classification.

3.3. Screening of key water quality indicators based on SHAP importance ranking

To interpret the decision-making process of the Auto-sklearn model, a SHAP analysis was performed on the single representative run (selected based on its Weighted F1 being closest to the five-run average). The analysis quantified the impact of each water quality indicator on class predictions, with each point in [Fig. 6](#) representing a Shapley value where color indicates the relative magnitude of the indicator value. For Class I waters ([Fig. 6a](#)), pH demonstrated the highest importance,

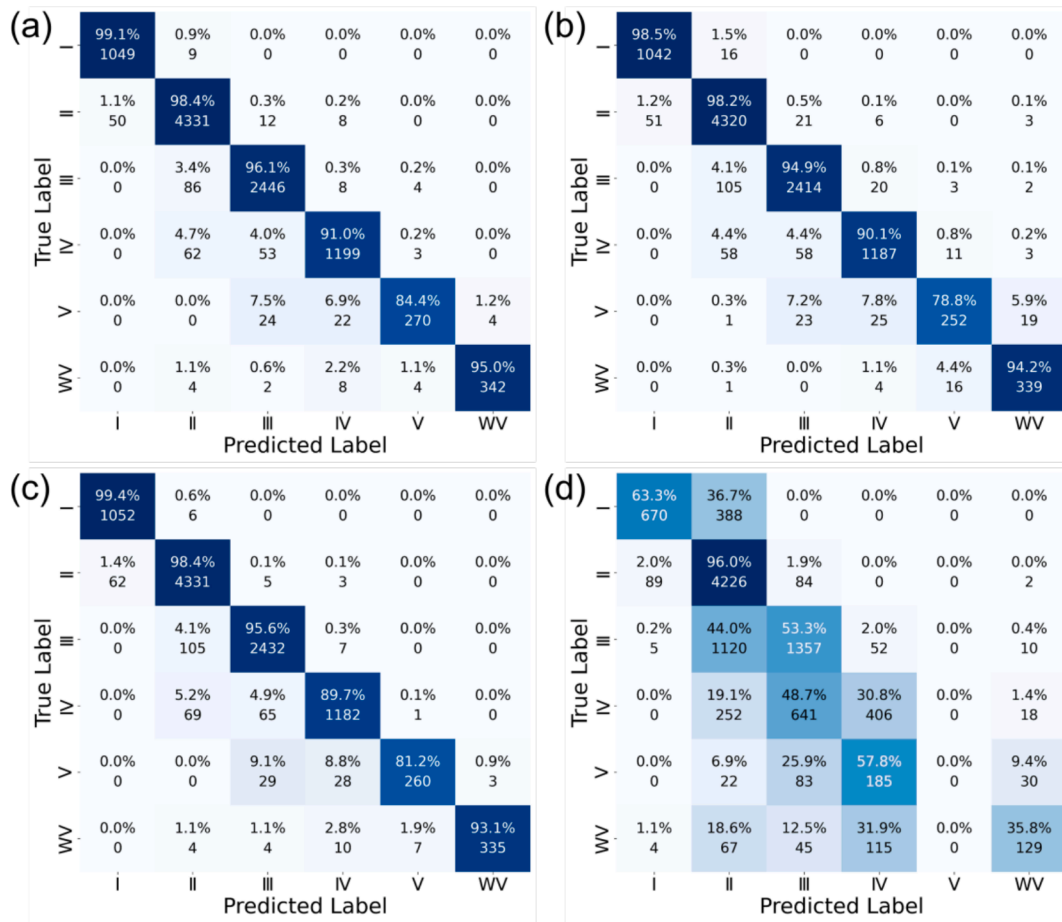


Fig. 4. Comparison of confusion matrices for water quality classification using different ML models: (a) Auto-sklearn, (b) XGB, (c) RF, and (d) LR. The results shown are from a representative run for each model.

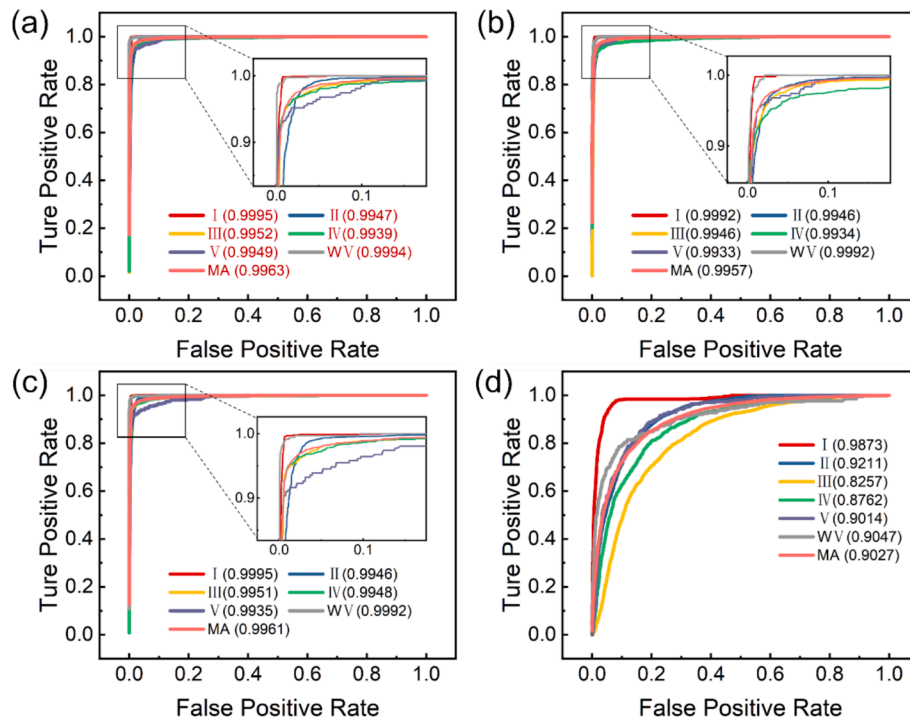


Fig. 5. ROC curves and AUC values for different water quality classification algorithms: (a) AutoML, (b) XGB, (c) RF, and (d) LR. Both the plotted curves and the corresponding AUC values are from a representative run for each model.

Table 2

Model performance (Weighted F1, Mean $\pm$ SD) across different validation strategies.

Model	Random split validation	Temporal validation	Spatial validation
Auto-sklearn	0.9633 $\pm$ 0.0027	0.9627 $\pm$ 0.0005	0.9675 $\pm$ 0.0154
RF	0.9607 $\pm$ 0.0029	0.9612 $\pm$ 0.0018	0.9670 $\pm$ 0.0157
CatB	0.9596 $\pm$ 0.0027	0.9595 $\pm$ 0.0004	0.9620 $\pm$ 0.0155
DT	0.9566 $\pm$ 0.0024	0.9572 $\pm$ 0.0026	0.9640 $\pm$ 0.0160
LGBM	0.9560 $\pm$ 0.0035	0.9563 $\pm$ 0.0011	0.9577 $\pm$ 0.0146
XGB	0.9545 $\pm$ 0.0030	0.9533 $\pm$ 0.0006	0.9544 $\pm$ 0.0148
KNN	0.8492 $\pm$ 0.0023	0.8453 $\pm$ 0.0036	0.8337 $\pm$ 0.0138
LR	0.6457 $\pm$ 0.0026	0.6427 $\pm$ 0.0031	0.6453 $\pm$ 0.0610

followed by DO and TP, where higher pH values (red points) significantly contributed to Class I classification. In Class II (Fig. 6b), TP emerged as the dominant indicator, with COD_{Mn} and NH₃-N showing secondary importance.

For intermediate classes (III and IV), COD_{Mn} exhibited the strongest influence, with notably negative SHAP values for high indicator concentrations (Figs. 6c, d). The impact of TP and DO are showed varying patterns between these classes, suggesting their role in fine-tuning classification decisions. Class V and WV predictions (Figs. 6e, f) were primarily driven by COD_{Mn}, DO, and TP, with their extreme values being particularly indicative of degraded water quality. Across all classes, TP, COD_{Mn}, and DO consistently demonstrate high importance, while indicators such as WT, NTU, and EC showed relatively minor contributions to the classification decisions. These SHAP importance rankings align

logically with the single-factor classification standard (Table S1). The SHAP analysis effectively quantifies which indicators most frequently serve as the “limiting factor” in the official classification rule. For instance, the high importance of COD_{Mn} for predicting poorer water quality classes (e.g., Class V, WV) is expected. This finding aligns perfectly with the national standard (Table S1), in which high COD_{Mn} values are a key condition that defines these poor water quality classifications.

A comprehensive importance ranking of the water quality indicators was established (Fig. 7a). This was achieved by first calculating the mean absolute SHAP values for each indicator within each of the five independent runs. These values were then normalized within each run to sum to one. The final importance scores shown are the average of these five normalized values. A one-way ANOVA with Tukey’s post-hoc test was performed on these final scores to assess statistical significance ($p < 0.05$). As denoted by the letters on the chart, the analysis identified COD_{Mn} and TP as the most influential indicators; their importances were not statistically different from each other but were significantly greater than all other indicators. DO ranked third, representing the next distinct level of statistical importance. A detailed visualization of the cumulative importance for each indicator, broken down by class, is provided in the supplementary materials (Fig. S6).

To screen for the key water quality indicators, we progressively incorporated them into the Auto-sklearn model according to the importance ranking. As shown in Fig. 7b, model performance (Weighted F1) improved rapidly, reaching 0.9205 ± 0.0097 with the top three indicators (COD_{Mn}, TP, and DO). Crucially, the addition of the fourth indicator (NH₃-N) did not yield a statistically significant improvement in model performance ($p > 0.05$), which is visually supported by the

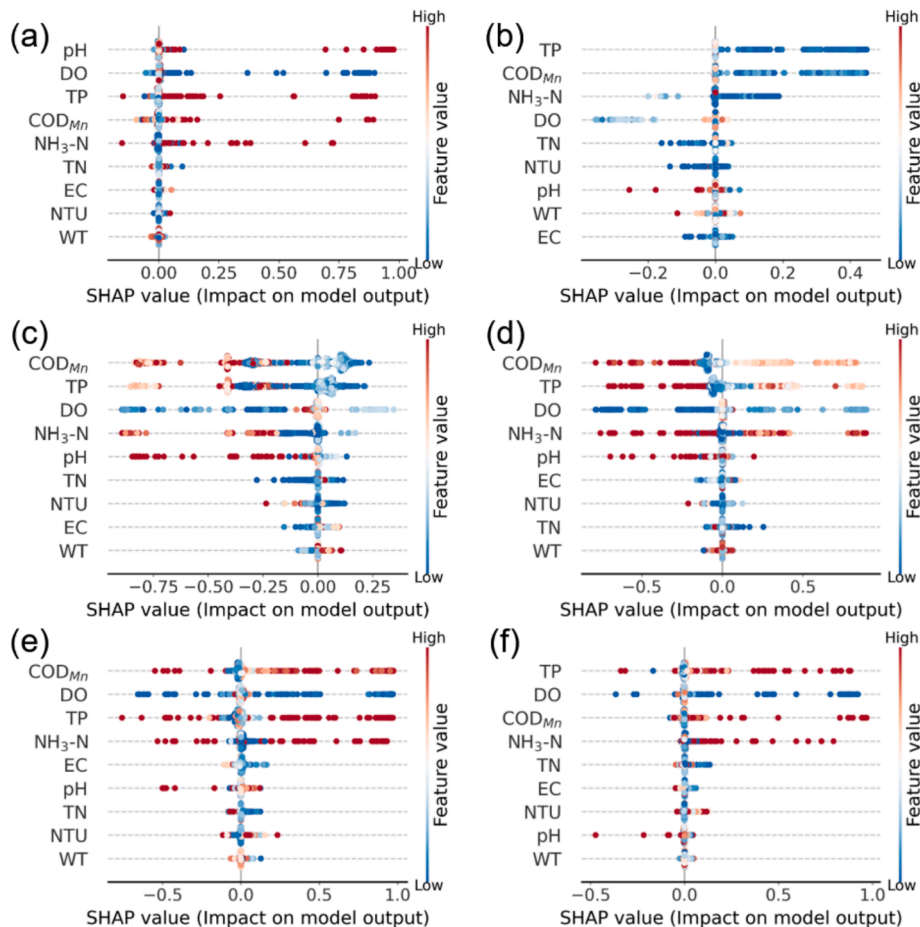


Fig. 6. SHAP importance plots showing the contribution of different water quality indicators to each water quality classes prediction: (a) Class I, (b) Class II, (c) Class III, (d) Class IV, (e) Class V, and (f) Class WV. These plots were generated from the representative run of the Auto-sklearn model.

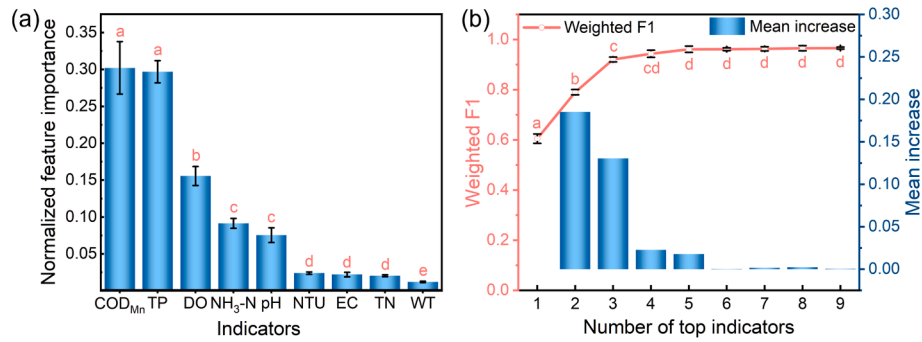


Fig. 7. Screening of key water quality indicators. (a) Overall indicator importance ranking showing the five-run mean of normalized SHAP values (where values within each run were normalized to sum to one). (b) Model performance evolution (Weighted F1, red line) and the incremental F1 increase (blue bars) as top-ranked indicators are sequentially added to the model. Different letters denote statistically significant differences (Tukey's HSD, $p < 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

significance letters on the chart. The mean Weighted F1 increase, as shown by the blue bars, was most substantial for the second (0.1851) and third indicators (0.1305). This lack of significant improvement beyond the top three provides strong evidence that COD_{Mn}, TP, and DO can serve as a core set for efficient and cost-effective water quality monitoring at a national scale.

3.4. Regional variations in key water quality indicators across major basins

Building upon the identification of COD_{Mn}, TP, and DO as the key water quality indicators at the national scale, the SHAP analysis was extended to evaluate regional variations. To ensure the robustness of these regional findings, the analysis for each basin was performed using five independent model runs with different random seeds. The final importance scores, representing the mean of the five normalized SHAP values, are visualized in Fig. 8. The detailed results, including the mean and SD for each indicator in every basin, are available in the [supplementary material](#) (Table S9).

The results revealed a high degree of consistency across most of China. Specifically, seven of the nine major river basins—the

Continental, Hai, Huai, Pearl, Southeast, Southwest, and Yangtze River Basins—retained COD_{Mn}, TP, and DO as their three most critical indicators, mirroring the national-level analysis. This widespread agreement across diverse geographical regions further validates the fundamental significance of these three indicators in water quality assessment.

However, notable regional variations emerged in the two northern basins. In both the Song-Liao and the Yellow River Basins, NH₃-N replaced DO as one of the top three key indicators (Table S9). These findings highlight that while COD_{Mn}, TP, and DO form an effective national minimum indicator set, regional hydro-environmental characteristics may necessitate tailored monitoring approaches.

4. Discussion

4.1. Advantages of Auto-sklearn in water quality classification

In this study, Auto-sklearn demonstrated leading performance in water quality classification compared to traditional ML models, achieving the highest scores across multiple evaluation metrics (Table 1). Through automated model selection and optimization, Auto-

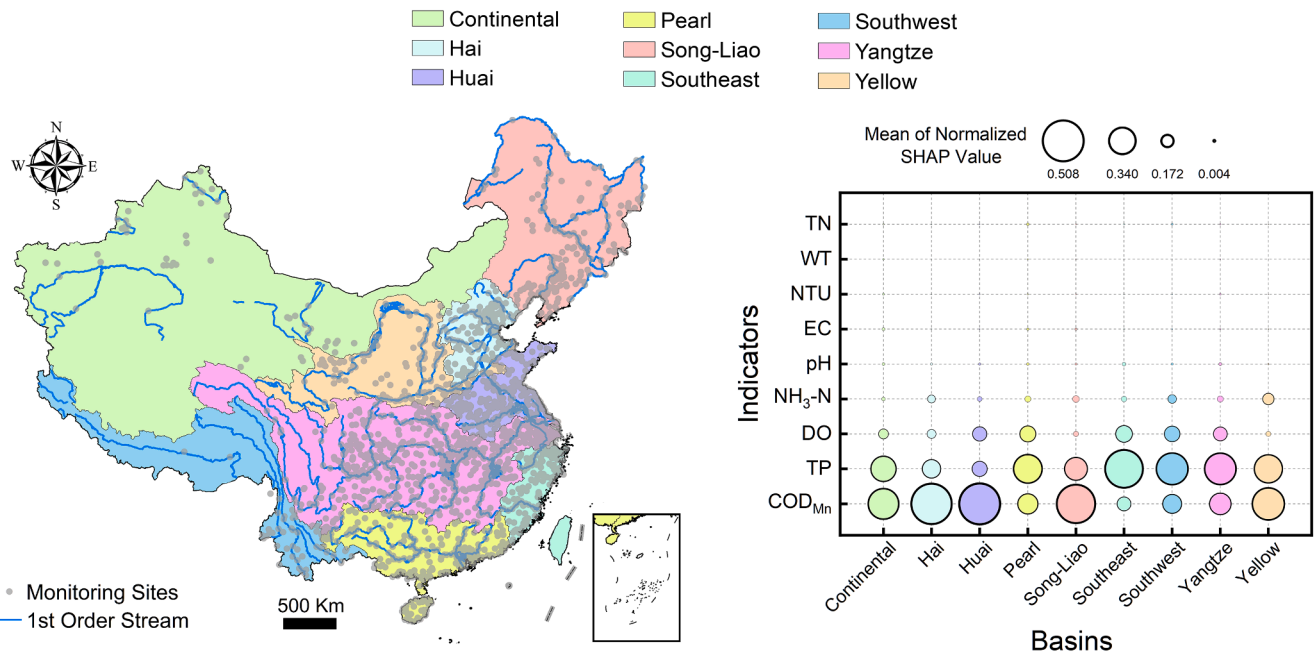


Fig. 8. Regional variations in key water quality indicators across major basins in China. The bubble size represents the relative importance of each indicator for water quality classification in different basins, quantified as the mean normalized SHAP value calculated from five independent runs.

sklearn successfully identified and combined complementary models into an ensemble of 7 ML pipelines (Table 1), automating the multi-stage manual workflow required by traditional ML approaches into an integrated process (Fig. 2). While we compared Auto-sklearn with various ML models both inside and outside the framework to validate its performance, it is important to note that conventional ML models theoretically cannot outperform AutoML since they are part of the model selection library of Auto-sklearn. This aligns with previous findings where even experienced data scientists could not surpass automated optimization of Auto-sklearn in specially designed challenge competitions (Guyon et al., 2019).

The performance improvement of Auto-sklearn over RF was relatively modest (Table 1), as RF was automatically selected as the optimal classifier across all seven ensemble pipelines (Table 1), which aligns with findings from previous studies in clinical diagnostic applications (Bifarin and Fernández, 2024). The consistent selection of RF validates the automated model search effectiveness of Auto-sklearn and indicates tree-based ensemble methods with moderate complexity work particularly well for classification tasks (Chan et al., 2023; Senocak et al., 2023).

All in all, AutoML frameworks such as Auto-sklearn are suggested to be employed directly for ML tasks in environmental monitoring instead of conducting extensive model screening and hyperparameter tuning. This approach not only ensures optimal model selection through automated processes but also saves considerable time typically spent on comparing various standalone algorithms. Although AutoML may require substantial computational resources, with the rapid advancement of computing infrastructure and increasing accessibility to high-performance computing resources, computational demands are becoming less of a constraint. The successful application in water quality classification demonstrated here provides a representative case for broader adoption of AutoML in environmental science research.

4.2. Rationality and practicality of key water quality indicators screening

4.2.1. A cost-effective nationwide indicator set

Water quality classification assessment traditionally relies on a comprehensive set of indicators, with national standards such as GB3838-2002 requiring monitoring of 24 fundamental indicators (MEEPRC, 2002). While this extensive monitoring approach ensures accuracy, it poses significant challenges in terms of cost and time efficiency (Sun et al., 2021). Our SHAP-based analysis identified COD_{Mn}, TP, and DO as the most influential indicators for water quality classification in China (Fig. 7a), offering a potential solution to this practical challenge.

The statistical effectiveness of this three-indicator set is robust. Our SHAP-based sequential analysis showed that these indicators alone could achieve a high Weighted F1 of 0.9205 ± 0.0097 , capturing the majority of information necessary for accurate classification. To validate this further, we compared the Auto-sklearn trained on these three indicators against a programmatic baseline applying the standard single-factor rule to the same limited set. As detailed in Table 3, the Auto-sklearn model achieved a statistically significantly higher Weighted F1-score (0.9205 ± 0.0097) compared to the rule-based baseline (0.8976 ± 0.0038). More importantly, the Auto-sklearn model demonstrated superior performance across all classes, most critically for Class WV, where it achieved an F1-score of 0.8285—a substantial 18.9 % relative improvement over the baseline's 0.6968. This superiority stems from the Auto-sklearn's ability to capture complex, non-linear interactions that the rigid rule inherently misses, confirming that the combination of our key indicators with an intelligent model provides a more accurate and practically valuable assessment, especially for managing the most polluted water bodies.

The identification of these key indicators provides a practical solution to the prohibitive costs associated with comprehensive water quality monitoring. By focusing on just three indicators (COD_{Mn}, TP, and

Table 3

Benchmarking of the Rule-Based Baseline against Auto-sklearn using the nationwide key indicator set (COD_{Mn}, TP, and DO), presented as mean $\pm$ SD.

	Rule-Based Baseline	Auto-sklearn
Precision	0.9030 ± 0.0036^a	0.9245 ± 0.0026^b
Recall	0.9017 ± 0.0037^a	0.9236 ± 0.0024^b
Weighted F1	0.8976 ± 0.0038^a	0.9205 ± 0.0156^b
Macro F1	0.8498 ± 0.0053^a	0.8885 ± 0.0078^b
Class I	0.9279 ± 0.0044^a	0.9447 ± 0.0025^b
Class II	0.9255 ± 0.0035^a	0.9429 ± 0.0026^b
Class III	0.8939 ± 0.0049^a	0.9142 ± 0.0052^b
Class IV	0.8645 ± 0.0091^a	0.8878 ± 0.0015^b
Class V	0.7902 ± 0.0182^a	0.8128 ± 0.0217^b
Class WV	0.6968 ± 0.0077^a	0.8285 ± 0.0097^b

Note: Different letters in superscript indicate a statistically significant difference between the Rule-Based Baseline and the 3-indicator Auto-sklearn model (Tukey's HSD, $p < 0.05$).

DO), monitoring programs can achieve reliable assessment while reducing both capital investment for equipment and operational expenses for chemical analysis. This simplified approach is particularly valuable for developing regions with resource constraints, making systematic water quality monitoring more economically feasible and widely implementable.

4.2.2. Regional heterogeneity

While our reduced indicator set (COD_{Mn}, TP, and DO) performs well for general classification at the national scale, our regional analysis critically demonstrates that optimal indicator combinations vary in response to distinct, regional-scale anthropogenic pressures. As shown in Fig. 8 and Table S9, the Song-Liao and Yellow River Basins required a different key indicator set where NH₃-N replaced DO. This shift is a direct reflection of the region-specific pollution drivers.

The exceptional severity of nitrogen pollution in these basins is quantitatively supported. The Song-Liao River Basin, for instance, has a Net Anthropogenic Nitrogen Input estimated at a staggering 9453 kg N km⁻² yr⁻¹, a value noted to be higher than that of southern basins (Cao et al., 2025). Similarly, the Yellow River Basin is the most severe region for agricultural non-point source pollution in China, with its provinces contributing a disproportionately high 42.67 % of the entire nation's agricultural TN emissions (Tan et al., 2025). This extreme loading means that raw pollutants, rich in NH₃-N, are the dominant feature of the pollution profile.

This immense pollution pressure is critically amplified by the region's distinct hydrological conditions, which starkly contrast with other major basins. The northern basins are characterized by extremely low water availability. The annual average runoff of the Liaohe (36.7 km³) and Yellow (52.1 km³) rivers is an order of magnitude lower than that of the southern Yangtze and Pearl rivers, which both exceed 300 km³ (Gao et al., 2020). The Yellow River exemplifies this stress, irrigating 13 % of the nation's cropland with only 3 % of its total river runoff (Ren et al., 2016). This "high load, low flow" dynamic means that the immense nitrogen inputs are concentrated in a minimal volume of water, leading to exceptionally high pollutant concentrations that are not seen in high-flow southern basins.

This comparative analysis confirms that the nitrogen pollution in the Song-Liao and Yellow River Basins is uniquely prominent. While southern basins like the Yangtze may have large total nitrogen loads, their vast dilution capacity mitigates the concentration and, thus, the immediate severity (Wang et al., 2020). In contrast, the combination of extreme nitrogen input intensity and severely limited dilution capacity in the Song-Liao and Yellow River Basins creates an environment where primary pollutants like NH₃-N are a more direct and sensitive signal of degradation than secondary ecosystem responses like DO. This evidence validates the findings of our model and underscores the necessity of basin-specific indicator sets for effective water quality management.

4.3. Limitations and future perspectives

Several limitations of this study should be acknowledged from both technical and methodological perspectives. Firstly, the indicator optimization in this study was confined to only nine commonly monitored indicators, which represents a subset of the comprehensive indicator system required by Chinese national standards (Table S1). Many important indicators specified in the standards, such as heavy metals, volatile phenols, and petroleum, were not included in our analysis. Moreover, biological indicators like Chlorophyll-a and algal density had to be excluded due to significant missing values in the monitoring data. These data availability constraints mean that our key indicator screening results might not fully reflect all aspects of water quality assessment required by national standards, particularly in terms of biological water quality and specific pollutants. To address this limitation, future studies should expand the scope of indicator analysis to include all indicators specified in national standards, which would provide a more comprehensive basis for key indicator selection.

Secondly, while our current AutoML framework successfully reduces the dimensionality of water quality indicators, it can only perform classification when all key indicators (COD_{Mn}, TP, and DO) are available. The model becomes inoperable when any of these key indicators are missing due to monitoring equipment malfunction or data transmission issues, which is a common scenario in environmental monitoring. To address this temporal missing data challenge, future research should incorporate time series models (e.g., RNNs, Transformers) to first predict missing key water quality indicators using their historical measurements, and then feed these predictions into the AutoML classification framework. This two-stage approach would enable continuous water quality classification even during periods of missing measurements, further enhancing the practical applicability of the system.

Despite these limitations, the framework presented in this study offers significant practical implications. Our work provides a data-driven pathway to design more cost-effective monitoring networks, first by identifying a generalizable set of core indicators (COD_{Mn}, TP, and DO) at the national scale. Crucially, the framework also demonstrates its ability to adapt to local conditions by identifying different key indicators (e.g., NH₃-N) in basins with unique pollution pressures. This adaptability is particularly valuable for regions with limited financial or technical resources. Moreover, the integration of an AutoML framework automates the otherwise complex and labor-intensive process of model selection and hyperparameter tuning, lowering the technical barrier for environmental agencies to implement predictive analytics. Ultimately, this study provides a scalable and adaptable template for creating more efficient and intelligent water quality management systems, enabling a more strategic allocation of resources in environmental protection.

5. Conclusion

This study demonstrates the effectiveness of integrating Auto-sklearn with interpretable ML techniques for water quality monitoring and classification optimization. Auto-sklearn outperformed most traditional machine learning approaches in water quality classification, achieving the top-tier performance with a Weighted F1 of 0.9633 ± 0.0027 . Crucially, the framework exhibited robust and generalizable performance, maintaining its superiority across rigorous temporal and spatial validation schemes. Furthermore, SHAP analysis revealed COD_{Mn}, TP, and DO as the most influential indicators. Crucially, a model using only these three indicators achieved a high Weighted F1-score (0.9205 ± 0.0097) and outperformed the rule-based baseline, yielding higher F1-scores across all classes and a 18.9 % relative improvement for Class WV water, offering potential for streamlined monitoring strategies. Regional analysis demonstrated that while most river basins aligned with the nationwide key indicator set, spatial heterogeneity exists in Song-Liao and Yellow River Basin where NH₃-N emerged as a key indicator.

The proposed framework addresses two critical challenges in water quality assessment: it eliminates the labor-intensive process of traditional ML implementation through automated workflow of Auto-sklearn, while SHAP-based indicator screening substantially reduces monitoring costs by identifying key indicators. This integration provides a practical solution for water environment management, particularly valuable for developing regions where both technical expertise and monitoring resources are limited.

CRediT authorship contribution statement

Wenjie Qin: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Heng Zhong:** Writing – review & editing. **Jingping Hu:** Writing – review & editing, Funding acquisition. **Wei Luo:** Investigation. **Zhe Shao:** Validation. **Huijie Hou:** Writing – review & editing, Funding acquisition. **Zhenzhen Liu:** Data curation. **Xiaotong Zhang:** Validation. **Sha Liang:** Validation. **Jiakuan Yang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2025.134634>.

Data availability

Data will be made available on request.

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